EPSR-test-ex43 writexyz with pre-existing output files

Program(s): writexyz | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises the file-handling branches of `writexyz` [1]: scanning an existing append file block by block (including an incomplete last block and trailing blank lines), continuing a numbered series from an existing `.nblock` counter, a centre molecule beyond the number of molecules, and an exclusion list ended by a non-integer word.

Code exercised

Fortran (original, fortran_src/):

- `writexyz.f`: `run_writexyz` (command-line words / stdin prompts), `writexyz` (selection, rotation, block counting, append / numbered files)
- `ato_read_and_save.f`: `minitc` (reading the `.ato`); `minimagec.f`: `minimage` (minimum image about the centre molecule)
- `ato_manipulation_rev4.f`: `centreofmass`, `rotatecoords` (Euler rotation)

Rust (port, rust/epsr-core/src/):

- `tools/writexyz.rs`: `run`, `writexyz`, `block_atoms`
- `ato.rs`: `State::minitc`; `neighbour.rs`: `minimage`
- `tools/groupc_common.rs`: `centreofmass`, `rotatecoords`

Physical system

Configurations of the EPSR-test DMSO/water example: single `DMSO.ato` (10 atoms) and `WaterMolecule.ato` (3 atoms) templates, and the 200 DMSO + 400 water box `200DMSO400Water.ato` (3200 atoms, cubic 33.1213 Å, 300 K). Transformations only: Euler rotations (0°, 90°, 0°), (10°, 0°, 0°) and (-20°, 30°, -40°); slab |x| <= 6, |y| <= 7, -8 <= z <= 9 Å about the origin for the box.

Inputs

Pre-existing files placed in the case folder: `DMSOappend.xyz` (one complete block followed by an incomplete one), `WaterMoleculeappend.xyz` (two complete blocks and extra blank lines), `200DMSO400Water.nblock` containing 7. Commands: `writexyz DMSO.ato append 1 5 5 -5 5 0 90 0 0`, `writexyz WaterMolecule.ato append 0 50 50 -50 50 10 0 0 0`, `writexyz 200DMSO400Water.ato noappend 999 6 7 -8 9 -20 30 -40 4 abc 3`.

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Files compared: `DMSOappend.xyz` (the Fortran rewinds on the incomplete block, so the file is rewritten with block 00001 only), `WaterMoleculeappend.xyz` (block 00003 appended, blank lines kept), `200DMSO400Water00008.xyz` (168 atoms; centre = origin since 999 > nmol; type 4 excluded, type 3 not because the list stops at `abc`) and the updated `.nblock` (8).

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.